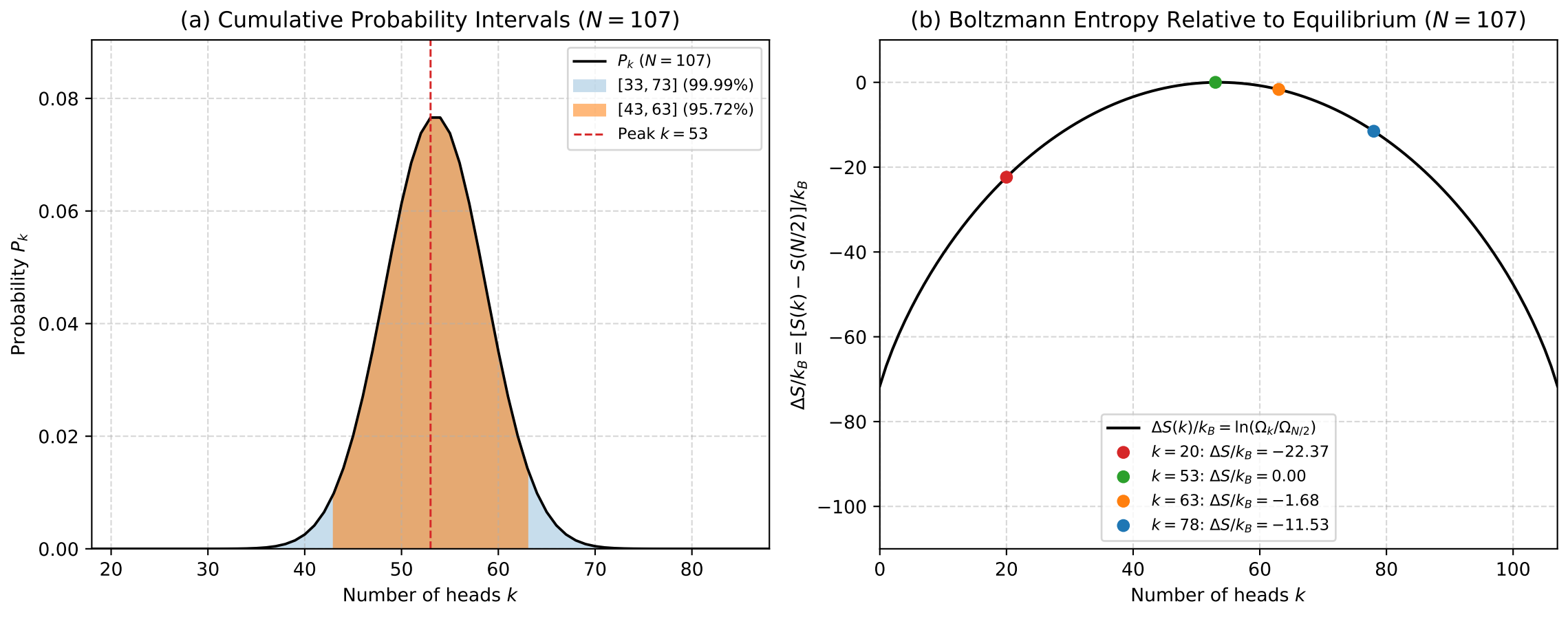
(a) Cumulative Probability Intervals ($N = 107$)
P_k ($N = 107$)
[33, 73] (99.99%)
[43, 63] (95.72%)
Peak $k = 53$
Probability P_k
Number of heads k
(b) Boltzmann Entropy Relative to Equilibrium ($N = 107$)
$\Delta S/k_B = [S(k) - S(N/2)]/k_B$
$\Delta S(k)/k_B = \ln(\Omega_k/\Omega_{N/2})$
$k = 20$: $\Delta S/k_B = -22.37$
$k = 53$: $\Delta S/k_B = 0.00$
$k = 63$: $\Delta S/k_B = -1.68$
$k = 78$: $\Delta S/k_B = -11.53$
Number of heads k